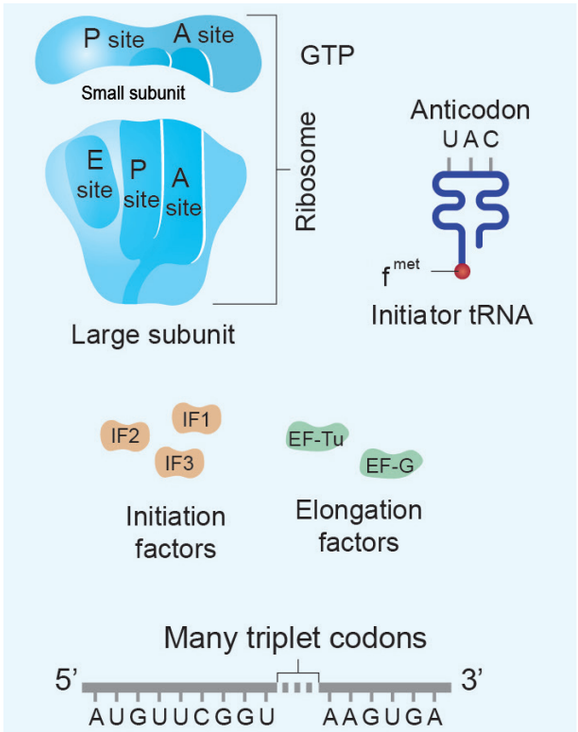


Fig. 5.13 a-Translation components

or **S-D sequence**. This sequences base-pairs with a region of the 16Sr RNA of the small ribosomal subunit facilitating initiation. The subunits of the ribosomes (30 S and 50 S) are usually dissociated from each other when not involved in translation **(Fig. 5.13a)**.

Initiation of translation in *E. coli* begins with the formation of an initiation complex, consisting of the 30S subunits of the ribosome, a messenger RNA and the charged N-formyl methionine tRNA (f^{met} – t RNA f^{met}), three proteinaceous initiation factors (IF1, IF2, IF3), GTP(Guanine Tri Phosphate) and Mg^{2+}.

The components that form the initiation complex interact in a series of steps. IF3 binds to the 30S and allows the 30S subunit to bind to mRNA. Another initiation protein (IF2) then enhances the binding of charged formyl methionine tRNA to the small subunit in response to the AUG triplet. This step 'sets' the reading frame so that all subsequent groups of three ribonucleotides are translated accurately.

The assembly of ribosomal subunits, mRNA and tRNA represent the initiation complex. Once **initiation complex** has been assembled, IF3 is released and allows the initiation complex to combine with the 50S ribosomal subunit to form the complete ribosome (70S). In this process a molecule of GTP is hydrolyzed providing the required energy

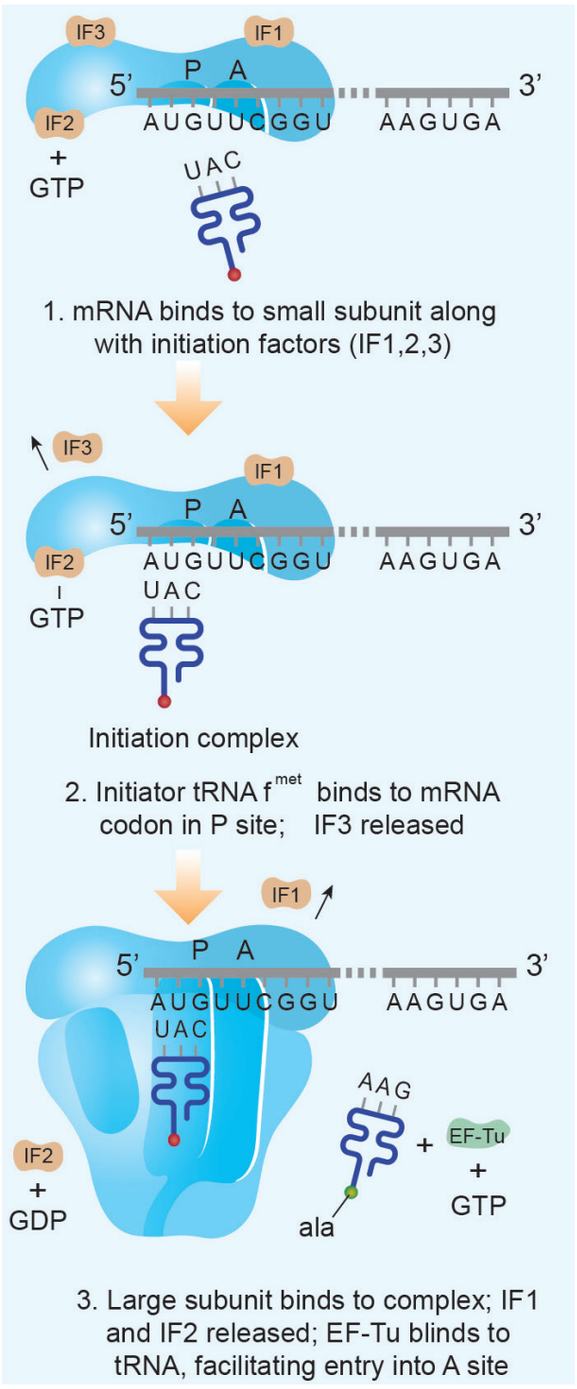


Fig. 5.13 b- Initiation

XII Std Biology-Zoology Chapter-5 EM.indd 76 23/12/2022 12:16:02